

Demonstration of VRBubble: Enhancing Peripheral Avatar Awareness for People with Visual Impairments in Social Virtual Reality

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ABSTRACT

Social Virtual Reality (VR) use is growing for socializing and collaboration. However, current applications are not accessible to people with visual impairments (PVI) due to their focus on visual experiences. We aim to design VR technologies to enhance social VR accessibility for PVI. We focus on facilitating peripheral awareness, a vital ability in social activities. With an iterative design process involving five participants, we designed *VRBubble*, a VR technique to facilitate peripheral awareness for PVI via spatial audio. Based on Hall's proxemic theory, *VRBubble* divides the social space with three Bubbles—Intimate, Personal, and Social Bubble—generating spatial audio feedback to distinguish avatars in different bubbles and provide suitable avatar information. We provide three audio alternatives: earcons, verbal notifications, and sound effects. PVI can select and combine their preferred feedback alternatives for different social context to maintain avatar awareness in a dynamic social VR environment.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Accessibility technologies**.

KEYWORDS

visual impairments, social virtual reality, proxemics, audio feedback

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1 INTRODUCTION

Social virtual reality (VR), such as VRChat, Rec Room, or Altspace, refers to VR platforms that allow users to socialize with each other within a virtual space to imitate real-life gatherings such as parties

or conferences [9]. Compared to 2D video conferencing systems that cause "Zoom fatigue" [11], social VR offers an immersive and engaging experience that can enhance interpersonal interaction and boost productivity. Unfortunately, similar to most mainstream VR applications, current social VR mainly targets sighted users by providing various visual avatar design and supporting non-verbal social interactions, which poses barriers to people with visual impairments (PVI). It is important for PVI to have equal access to social VR as virtual collaboration and gathering increases, especially during the COVID-19 pandemic [13].

Researchers have started tackling the VR accessibility problems for PVI by enabling them to navigate and perceive VR scenes [7, 12, 23, 24]. Some research leveraged or created additional devices (e.g., PHANToM) to support VR navigation for PVI [3, 7, 8, 10, 18, 23]. For example, Jansson et al. developed a haptic virtual environment where PVI can navigate the environment using PHANToM devices [7]. In a similar vein, Zhao et al. created a custom wearable controller to enable blind users to explore VR scenes by simulating the white cane interactions [23].

Besides using custom hardware, some researchers focused more on software solutions, designing VR interactions based on existing VR setup, such as off-the-shelf VR controllers, or keyboard and mouse for a virtual world [12, 15, 17, 21, 22, 24]. For example, Trewin et al. presented environment objects as a menu, along with verbally described information about an object when selected, such as name or distance [22]. Andrade et al. [1] used echolocation to provide spatial knowledge of virtual spaces for PVI, where the user can produce mouth-click or clap sound by pressing a key on a keyboard and hear the sound reflection from the environment. However, the ecolocation method was only used by a small amount of blind people. Recently, Nair et al. designed a technique that allowed blind users to look around the virtual environment in 3D games by scrubbing the thumbstick on a game controller to different directions; the system then announced what were in that direction via spatial audio [12].

However, prior work has mainly focused on basic VR tasks such as navigation and object perception. It does not address the unique barriers caused by the dynamic and multiplayer nature of social VR. Unlike a static VR scene with only system-generated objects, social VR is more complex and challenging. Human controlled avatars constantly move in the environment and interact with other avatars and objects. This information is mainly presented visually in social VR and is not accessible to PVI. To our knowledge, no existing techniques have focused on the avatar dynamics in social VR to support accessible social interactions for PVI.

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We aim to fill this gap by enhancing PVI's awareness of surrounding avatars in social VR. Unlike prior work that required PVI to actively explore and query information from the environment [12, 23], we focus on *peripheral awareness*—the innate ability to unconsciously "maintain and constantly update a sense of our social and physical context" [16]. People can usually maintain peripheral awareness effortlessly without being distracted from their main focus [2]. This ability is especially important in social and collaborative environments since it provides more context for one's activity. For example, when navigating to a specific location, a sighted person can easily notice who has passed by to decide whether to greet that person or start a quick conversation; or when a sighted person is in the middle of a conversation, she can stay aware of who else just joined the conversation or who is close enough to overhear the conversation.

We seek to facilitate peripheral awareness for PVI to better support them in social VR. We designed *VRBubble*, an audio-based VR technique that provides surrounding avatar information based on social distances. Following Hall's proxemic theory [6], we split the social space with three bubble layers centered on the user; the Intimate Bubble, Personal Bubble, and Social Bubble—to distinguish avatars at different social distances that can potentially be available for corresponding social interactions. VRBubble then provides different audio alternatives (i.e., earcons, sound effects, and verbal notifications) to convey the avatar information and dynamics, such as names, relationships with the PVI, and their interactions with the bubbles (entering or leaving). PVI also have the flexibility to select and combine their preferred audio alternatives for different bubbles and avatars to maintain awareness of the social context. Using the method of user-centered design [14], we iterate our prototype via a formative study with five PVI, exploring their experiences with our initial prototype in different social contexts (i.e., navigation, conversation) and improving our design accordingly. We describe the design iterations of VRBubble and the formative study in this paper. We envision that our work will make social VR more accessible to PVI and provide insights for future VR accessibility.

2 INITIAL PROTOTYPE

Our initial design of VRBubble follows the design considerations (C1-C3) below.

(C1) *Leverage mainstream platforms.* We focus on desktop VR, where a user can see the virtual environment on the desktop screen, hear the audio feedback via earphones or speakers, and interact with it via keyboard and mouse. While VR headsets are emerging, most people don't own a headset due to its cost [19], let alone PVI who cannot benefit from the visual feedback from the headset. Instead, desktop computers are more widely used and many social VR platforms (e.g., VRChat, AltSpace) support desktop access. Moreover, desktop VR with keyboard can support more comfortable and efficient interaction for longer use and higher productivity in social and collaborative context.

(C2) *Convey proper information about avatar dynamics.* Prior research has explored PVI's information needs for real-world social activities, and the top three include the identity of surrounding people, their relative location, and physical attributes [25]. So needs that translate equally to the virtual world are considered.

(C3) *Avoid intrusiveness and distraction.* Since peripheral awareness is supposed to be effortless, we seek to minimize the distraction of our design. We thus consider short but intuitive audio feedback at suitable timing to convey avatar information to PVI.

2.1 VRBubble based on Hall's Proxemic Theory

To inform PVI of the surrounding avatar dynamic without non-stop notifications, we follow Hall's proxemic theory [6] to divide the virtual environment into different social spaces. The theory correlates physical distances with social interactions that typically happen within that distance, such as distances taken for intimate interactions versus distances taken for friendly conversation. Three distances are defined: *intimate distance* (1 foot) where people usually feel distress when this space is encroached upon unwillingly, *personal distance* (4 feet) where people interact with familiar people, and *social distance* (12 feet) where conversation with less familiar acquaintances or group happens. The space outside of the social distance is considered to be public space.

Based on the proxemic theory, we define three levels of bubbles based on Hall's proxemic distances, including intimate, personal, and social bubbles (Figure 1). We then provide spatial audio feedback to convey the surrounding avatars' moving dynamic between these bubbles, thus enabling PVI to build suitable social interactions upon sufficient avatar awareness. Specifically, we use earcons to notify avatars' movement crossing different bubbles (i.e., enter, leave) and use verbal notification to provide more avatar information (e.g., name, relationship with the PVI). We describe our design for each bubble.

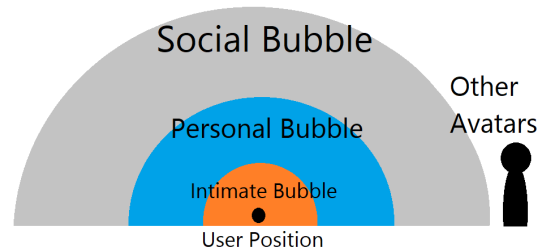


Figure 1: Conceptual Diagram of Bubbles

Intimate Bubble defines the space within the intimate distance. Since PVI cannot visually perceive whether they are too close to others or whether other people are invading their intimate space, we generate an spatial earcon that sounds like they are bumping into someone, ★, to alert them when avatars enter this bubble. We also add a short verbal notification "Collided with <Friend/stranger><name>" to provide the avatar identity, including name and relationship with the PVI (Figure 2(a)).

Personal Bubble represents the space between the intimate and personal distance. We create this bubble to inform the PVI when an avatar is close enough to hold a conversation with. This can also alert the PVI if someone is close to overhear her ongoing conversation to protect her privacy. We also notify the PVI when a certain avatar leaves the bubble to indicate that this avatar is no longer available for a personal conversation. A spatial two-beat earcon with increasing tone, ♪, is used to represent an avatar entering the



Figure 2: VRBubble Initial Design: (a) Intimate Bubble, (b) Personal Bubble, (c) Social Bubble.

bubble, and a two-beat earcon with decreasing tone, ♪ , for leaving. We also add a short verbal notification after the earcon to provide the avatar identity information: "<Friend/stranger> <name> is (or is no longer) nearby>." For example, if an avatar friend Alice enters the personal bubble, the PVI will hear, " ♪ , Friend Alice is nearby" (Figure 2(b)).

Social Bubble defines the space between personal and social distances. We generate audio feedback when an avatar enters this space, so that PVI would know that this avatar could be potential conversational partners, but would not be in the immediate distance to start the conversation. The PVI could then decide whether they are interested in approaching this person for a conversation. We use a spatial earcon with a timbre difference from that of the personal bubble earcon, with an increasing tone, ♪ , for entering the bubble and a decreasing tone, ♪ , for leaving. A verbal notification goes after the earcon to describe the avatar: "<Friend/stranger><name> is (or is no longer) in the vicinity" (Figure 2(c)).

Since audio notification can be distracting in general, we allow PVI to manually stop any ongoing audio feedback by pressing a Ctrl key (the same key was used in screen readers). For avatars outside of the Social Bubble, we regard them to be in the public space. We don't provide information about these avatars to reduce distraction.

2.2 Features for basic functionality

We added two features to enable PVI to complete basic movement in a virtual environment: (1) *Movement*: PVI can control the movement of their own avatar via arrow keys, for example, up arrow to move forward and left arrow to move left. Each key press would move the PVI a foot in the virtual space and play a footstep sound to notify the PVI. (2) *Turning*: Pressing or holding the left or right arrow keys while holding the shift key would turn the PVI in the corresponding direction. A ticking sound would play at regular intervals during the turning process, providing a sense of how much the PVI had turned.

3 FORMATIVE STUDY

Following the method of user-centered design [14], we conducted a formative study with five PVI to explore their experiences and feedback, thus improving our prototype based on participants' suggestions.

3.1 Method

3.1.1 Participants. We recruited five visually impaired participants (two male, two female, and one who preferred not to say) whose ages ranged from 39 to 58 ($mean = 48.6$, $SD = 7.76$, Table 1). The

participants were recruited through the National Federation of the Blind.

3.1.2 Apparatus. We implemented the initial design of VRBubble and a baseline in a VR environment. The baseline is an audio beacon that consisted of a spatial beeping sound, in intervals, attached to each avatar (except for the participant avatar) in the environment. We chose this baseline since audio beacons are an effective and common way of conveying the presence of objects for PVI [4].

We built the VR environment and the VRBubble and baseline features with A-Frame [20], a web framework that generates 3D spaces via HTML and Javascript. We hosted our prototype with Glitch [5] to enable a remote setup for the study. Participants and researchers can go to the same web address to join the same VR environment. Once connected, researchers can input unique command sequences via a keyboard to adjust the participant's client, such as changing from VRBubble features to baseline features or moving their avatar to a new location in the VR environment. Participants' behavioral data (e.g., their positions at different times) was then logged and output into the researchers' browser console. We asked participants to share their screen via Zoom if possible to ensure the commands from the researchers were properly reflected on the participants' end.

3.1.3 Procedure. Our study consisted of a single session that lasted 2 hours for each participant. We conducted the study remotely via a Zoom video call. An initial interview was first conducted to ask about participants' prior experience with VR, especially social VR. Then we asked about their social experiences in real life.

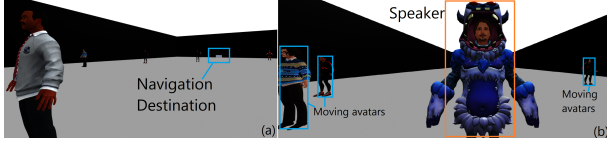
We then asked participants to experience our initial prototype (Section 2). Participants were requested to use their earphones for the study, since we generate spatial audio. We sent a url to our VR environment through email before the study. If any participant had difficulty accessing their email, we sent the link again through the Zoom chat or read the link out-loud to them.

After the participant successfully joined the VR environment, we provided a tutorial on how to use the baseline and VRBubble. Both design were presented as anonymous features to prevent bias. Participants then used our prototypes for two social VR tasks to simulate different social contexts, including (1) a navigation context: navigating as fast as possible to a destination marked with a spatial audio beacon distinct from VRBubble and baseline sounds (Figure 3(a)); and (2) a conversation context: listening to a short story told by an avatar and later answering questions about the story (Figure 3(b)). For each social context, avatars were rendered randomly around the participants.

We asked participants to complete the primary social task (i.e., navigation, conversation perception) and also maintain awareness

Table 1: Demographic Information

Pseudonym	Age/Sex	Visual Condition	Prior VR Experience
V1	39/Agender	Blind for 9-10 years	Microsoft Flight Simulator
V2	51/M	Legally blind since birth	Simulated roller coaster
V3	58/F	Blind since birth	Tried on a couple of VR headsets, explored a virtual world, made an avatar
V4	55/M	Blind since birth	Used many VR programs
V5	40/F	Blind since birth	Created avatars in some online video games

**Figure 3: Formative Study Tasks: (a) Navigation Context, (b) Conversation Context**

of the surrounding avatars, memorizing how many avatars have passed by and how many of them are friends. Participants were asked to prioritize the primary social task if they could not complete both the social and the awareness tasks at the same time. Participants performed each social task in two conditions: using VRBubble and using the baseline (one trial per condition). We altered the virtual destination and the story for each condition in the corresponding social context. We also counterbalanced the order of the two conditions across all participants to avoid order effects. After the trials in each social context, we asked participants about their feedback for VRBubble and the baseline. At the end of the study, we promoted the participants to brainstorm about what other designs can better help them with peripheral awareness of avatars.

3.2 Findings

3.2.1 General Experience with VRBubble. In general, everyone was fond of VRBubble. They liked that they could use the VRBubble as social boundaries, since each bubble represented different social ranges and provided more information than the baseline. Three participants (V1, V2, V5) preferred our bubble feature over the baseline.

However, participants found VRBubble distracting in the conversation context. Four participants (Except V2) mentioned having trouble concentrating on the conversation and the audio notifications from the surrounding avatars at the same time. As V4 indicated, "They were overlapping. If the story was a little louder, it might have been better, but the [notification from] avatars were louder than the story." Some participants (V1, V2, V3) wanted to be able to toggle off certain information when focusing on a conversation even if it meant less awareness towards surrounding avatars.

On the other hand, some participants (V2, V3) wanted more information if their social goal was to meet other avatars. They suggested being able to know the exact number of avatars in the room or having a short description of each avatar. V3 further explained the information she needed, "The main things like age, hair, hair style you know just kind of like their general appearance." To

better perceive the number of avatars in the room, V2 suggested having an audio notification for any avatar who enters the room even if they are not in the VRBubble.

We report participants' feedback on different audio feedback design below.

3.2.2 Collision Sound for Intimate Bubble. The design for Intimate Bubble (Section 2.1) was favored by most participants. V2 and V5 were intrigued by the ability of knowing that they collide with an avatar. As V2 mentioned, "I liked specially the noise where the person's right in my [intimate] space. I thought it's a good signal, even before [the collision] it tells you what it is."

3.2.3 Earcons for entering and leaving the bubbles. In general, all participants felt the earcons useful. V2, V4, and V5 found the earcon design intuitive to represent avatars' entering and leaving. V4 preferred the earcon over the verbal notifications, since it was the easiest way to focus on multiple avatars. V1 and V3 also liked that different earcons were used to distinguish different bubbles. However, some participants were worried about the learnability of the abstract earcons if there's no tutorial or instruction in a real VR application. As V3 indicated, "I don't know that I would know [the earcon] just on its own but once I knew that one was coming in and one was going out, it would make sense that, that's what they were for." Participants suggested having more intuitive real world sounds (e.g., a door chime for entering a bubble) instead of the abstract earcons.

Participants provided suggestions to improve the earcon design. First, participants wanted more flexibility to adjust the earcons, including speeding up/down the earcon notifications (V5) and toggling the earcons on and off (V2, V3, V4). Moreover, several participants (V1, V3, and V5) suggested adding new earcons to differentiate between friends and strangers when an avatar interacts with a bubble.

3.2.4 Verbal notifications. Most participants liked the verbal notifications, because it clearly alerted them to the avatar presence and explained some useful avatar information. Some participants (V2, V5) preferred the verbal notifications over other forms of feedback because it's easy to understand. V1 and V4 liked that the verbal description distinguished between friend and stranger avatars. Interestingly, they had different preferences in terms of prioritizing friends or strangers, depending on who they currently need information on. As V4 mentioned, "[The user] may want to automatically turn off certain players or certain avatars. Some avatars you may not even be concerned with."



Figure 4: VRBubble Final Prototype: (a) Earcons at different bubbles, (b) Verbal Notifications, (c) Sound Effect.

Participants provided some suggestions for verbal descriptions. Instead of using the phrases "in vicinity" and "nearby" to distinguish the conversational and social bubbles, participants (V2, V3) suggested hearing more concrete information (e.g., numerical distances) to tell how far away avatars were from them. Additionally, participants wanted to be able to adjust the speed of the narrator (V2, V3, V5) or mute the notification if needed (V2, V4, V5).

4 FINAL PROTOTYPE

4.1 New design considerations

We reflect on participants' feedback and distill new design considerations with corresponding prototype modifications.

Intuitive sound effects. While most participants found the earcon design useful, understanding the abstract earcons could create a steep learning curve for PVI. Compared to the abstract earcons, more intuitive audio feedback is needed. We thus add another audio alternative—sound effects that simulate the real-world sound (e.g., chatter sound, footsteps) to provide information of the peripheral avatars for PVI. This design could also potentially enhance the immersion of the social VR environment.

Sufficient information. Beyond avatar identity, participants wanted more avatar information, such as the number of avatars, descriptions of the avatars, and whether an avatar was a friend or a stranger via the earcon. In our next iteration, we provide more avatar information (e.g., physical attributes) about avatars while giving users the flexibility to select which information they need to avoid distraction (see Customization).

Customization. The study revealed PVI's need for flexibility and customization for different social tasks. For example, some preferred simpler information for the more distracting conversation task. Moreover, participants preferred different audio feedback alternatives: V4 liked the earcons, while V2 and V5 preferred the verbal notifications. We provide different customization options to enable PVI to customize VRBubble in the final prototype.

4.2 Final design of VRBubble

In the refined prototype, we provide three spatial audio feedback alternatives: earcons, verbal notifications, and sound effects from real world. For each alternative, we present more avatar information, including avatar dynamic between bubbles, avatar identity (name and/or relationship with the PVI), and avatar appearance. Instead of generating all audio feedback together, we allow PVI to select and combine different audio alternatives for different bubbles and different avatars (e.g., friend vs stranger) to customize their experiences. We also enable them to change the speed of the verbal notifications to meet their preferences. We describe the design of three audio alternatives for each bubble.

Earcon. Given that earcon is abstract and has a high learning curve, we used earcons to present simple information, such as distinguishing between different bubbles and whether the avatar is a friend. Instead of completely different earcons, we designed associated earcons with distinctions to minimize the learning curve. Similar to the initial design, we use a two-beat earcon with the tone of the last beat increased (or decreased) to indicate an avatar entering (or leaving) the social bubble. For the personal bubble, since avatars in this bubble probably requires more immediate attention, we use four-beat earcons with the tone of last beat increased or decreased for the interactions with this bubble, ♪♪♪♪ and ♪♪♪♪ . To distinguish friend and stranger avatars, we adjust the pitch and speed of the earcons, so that higher pitched, faster earcons indicate friends, while normal pitch and speed for strangers. The bump earcon for the intimate bubble was maintained.

Verbal Notifications. We provide more clear verbal notifications with more avatar information. Instead of using "nearby" and "in vicinity," we use "within speaking distance" and "within social space" to better clarify which bubble an avatar enters. Moreover, we provide the option for the PVI to hear the appearance of a friend avatar. For example, if a friend Alice enters the personal bubble, the PVI will hear, "Friend Alice within speaking distance, in a panda avatar." We don't provide appearance description for strangers to minimize distraction.

Sound Effect from Real World. This design alternative focuses on providing realistic real-world sounds to intuitively support peripheral awareness and immersion. We use crowd sound effects as the background sound to represent the number of avatars in the social bubble. Three levels of crowd sound effects with increasing volume and densities indicate three levels of avatar amount: 1-5, 6-10, and more than 10 avatars. Cheering crowd sounds are also used to highlight friend avatars due to their higher importance. Moreover, we put more emphasis on the avatars in the personal bubble since the PVI are more likely to converse with them. A spatial footstep sound effect is regularly played for every avatar within the personal bubble. We generate different footsteps to match the avatar appearance. At the current stage, the footsteps only distinguish male, female, and non-human avatars. In the future, we hope that VR designers will consider design corresponding footsteps or sound effects for different avatars to foster avatar accessibility.

5 FUTURE WORK

As the next step, we will conduct a summative evaluation with PVI to collect their quantitative performance using VRBubble as well as their qualitative preferences for different audio alternatives in different social context. Moreover, while VRBubble allows customization, we have not designed specific VR interactions to support that. We

will thus improve VRBubble by designing accessible and easy-to-use VR interaction techniques for PVI to enable the customization. We will also consider involving haptic feedback in VRBubble to improve PVI's experience.

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